Quantitative text analysis – Essex Summer School

Scaling methods, Wordscores, Wordfish, and LSS

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Today's class

- Learn how to use text data to estimate document positions on an underlying dimension through **scaling approaches**:
  - **Wordscores** (supervised)
  - **Wordfish** (unsupervised)
  - **Latent Semantic Scaling** (semi-supervised)
  - **Instruction-based scaling** (LLM-assisted)
- Lab session: scaling House of Commons leaders using text

Today's corpus: House of Commons leader speeches

- The lab uses a new House of Commons leader-period sample.
- Scaling models usually need longer documents than single short interventions.
- We therefore aggregate speeches into leader-election-period documents:
  - one document for each Labour or Conservative leader in each legislative period;
  - each document combines up to 200 speeches from that leader during that period;
  - if a party changes leader during a period, both leaders can appear as separate documents.
- Guiding question: what drives document placement? Do these models recover party ideology, historical period, agenda, role, style, or a mixture of these?

A scaling problem

Suppose we want to compare Labour and Conservative leaders discourse across legislative periods.

Research question	What we need to check
Do Labour and Conservative leaders separate?	Do estimated positions match the intended party contrast?
Do older and recent periods use systematically different language?	Are we recovering time rather than party ideology?
Do leadership changes within periods matter?	Are positions driven by party, period, leader, role, agenda, or style?

Keep in mind: a scaling model gives us a dimension; we still have to interpret and validate what that dimension means.

Motivation for scaling methods

- Research task: place documents or actors on a single ideological dimension
- Traditionally measured using:
  - Expert surveys (e.g. CHES)
  - Hand-coded content analysis (e.g. CMP)
  - Roll-call votes (parliaments)
- Limitations: expensive, labour-intensive, not always available
- **Idea: use the text that actors produce to estimate positions automatically**

Assumption behind text scaling methods

- Ideological dominance assumption (Grimmer & Stewart 2013)
- Word use varies systematically with ideological position
- E.g., **"death tax" vs. "estate tax"** in U.S. politics; "workers" vs. "business" in parliamentary language

  Republicans ... had to train members to say "death tax" ... Estate tax sounds like it only hits the wealthy but "death tax" sounds like it hits everyone. (Graetz & Shapiro 2005)

- Not all texts follow this logic → validate carefully!

Four broad approaches

Method	Type	Input needed
Wordscores	supervised	reference texts + scores
Wordfish	unsupervised	a DFM
LSS	semi-supervised	seed words + word embeddings
LLM scoring	instruction-based	scale definition + validation set

Evidence for each scaling method

Method	Where the direction comes from	What documents are scored by
Wordscores	reference documents with known dimension scores	words associated with each reference document
Wordfish	word-use differences in the corpus	the strongest latent pattern in word counts
LSS	researcher-chosen seed words	semantic proximity to seed words
Instruction-based scaling	a written scale definition	model interpretation of the text and prompt

The same corpus can therefore produce different scales, because each method listens for a different kind of signal.

Wordscores intuition

Wordscores starts from documents whose positions we claim to know.

reference texts $\rightarrow$ word scores $\rightarrow$ new document scores

- If a Labour reference text uses a word often, that word is pulled toward the Labour side of the scale. If a Conservative reference text uses it often, it is pulled toward the Conservative side.
- New documents are placed by the weighted average of the words they contain.
- In the lab, Neil Kinnock and Margaret Thatcher in 1983–1987 define the scale; other leader-period documents are predicted relative to those anchors.

Wordscores

The procedure follows these steps:

1. Identify ‘reference texts’ that represent the extremes of a political spectrum (e.g., left-right, liberal-conservative, government-opposition, pro-EU-anti-EU)
2. Assign reference values (numeric scores) to these texts (e.g., -1,1)
3. Each word in the reference texts is assigned a Wordscore, calculated based on the relative frequency of each word in these texts multiplied by the reference values.
4. Wordscores are then used to scale unseen texts by summing the product of all Wordscores and their relative frequencies in these texts.

Source: Laver, Benoit & Garry (2003)

Criticisms of Wordscores

See Bruinsma & Gemenis (2019) for detailed critique:

- Relies heavily on the choice of reference texts
- Can conflate style or genre with ideology
- Lacks a generative model for text (Lowe 2008)
- Reference scores may not transfer across time

Wordscores in quanteda

```r
library(quanteda)
library(quanteda.textmodels)

wordscores_anchor_period <- "48th Parliament (1983-1987)"
labour_anchor_leader <- "Neil Kinnock"
conservative_anchor_leader <- "Margaret Thatcher"

docvars(scaling_dfm, "reference_score") <- case_when(
    docvars(scaling_dfm, "leader") == labour_anchor_leader &
    docvars(scaling_dfm, "period") == wordscores_anchor_period ~ -1,
    docvars(scaling_dfm, "leader") == conservative_anchor_leader &
    docvars(scaling_dfm, "period") == wordscores_anchor_period ~ 1,
  TRUE ~ NA_real_
)

hoc_wordscores <- textmodel_wordscores(
  scaling_dfm,
  y = docvars(scaling_dfm, "reference_score"),
  scale = "linear",
  smooth = 1
)

hoc_wordscores_pred <- predict(hoc_wordscores, newdata = scaling_dfm)
```

Wordscores among selected Conservative and Labour leaders

leader-election-period document	Role	Wordscores
Labour Neil Kinnock 1983–1987	reference	-0.2677
Labour John Smith 1992–1997	virgin	-0.1588
Labour Margaret Beckett 1992–1997	virgin	-0.1543
Conservative Kemi Badenoch 2024–2025	virgin	-0.1512
Conservative Margaret Thatcher 1987–1992	virgin	-0.0438
Conservative John Major 1987–1992	virgin	-0.0282
Conservative Margaret Thatcher 1979–1983	virgin	0.1066
Conservative Margaret Thatcher 1983–1987	reference	0.2964

- Only the Kinnock and Thatcher 1983–1987 documents define the scale.
- The other documents are placed according to how much their language resembles either anchor.

Wordfish

- Wordfish also places documents on one dimension, but it does not ask us to provide reference texts or seed words.
- Unsupervised method – there are no reference texts.
- It looks for the strongest pattern of word-use differences in the DFM.
- That pattern may be ideological, but it may also be about period, party system, institutional role, agenda, or genre.
- We can orient the sign of the scale using known documents, but this does not make Wordfish supervised.

Interpretation is the hard part: Wordfish will estimate a scale even when the scale is not the one we expected.

Wordfish example: House of Commons

In the lab, we apply Wordfish to the House of Commons leader-election-period documents.

- 42 documents: one for each Labour or Conservative leader-period (up to 200 speeches)
- the DFM has 42 rows and 7,088 word-feature columns
- Wordfish estimates the strongest one-dimensional pattern in these word counts

Step	Object
1	party leader in an election period
2	aggregated speech document
3	row in the DFM
4	Wordfish position

Wordfish model: application

$y_{ij} \sim Poisson(\lambda_{ij})$ $\lambda_{ij} = exp(\alpha_i + \psi_j + \beta_j * \theta_i)$

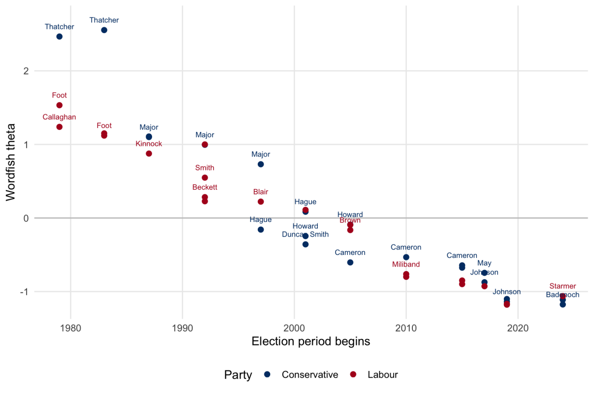
Applied to the House of Commons data, this models the number of times leader-election-period document i includes term j.

- Alpha α is a **document fixed effect**, controlling for some documents including more terms than others.
- Psi ψ is a **word fixed effect**, controlling for some terms being used more frequently than others.
- Beta β is the estimated **weight for each term used to position** the documents on the one-dimensional scale.
- Theta θ is the **estimate for each document on the one-dimensional scale**.

Wordfish model: documents

Estimated θ scores represent each document's position on the Wordfish dimension.

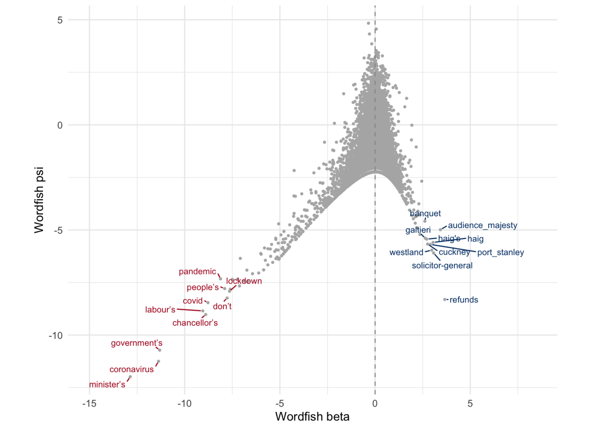
- The model is not told which party is left or right.
- It finds the strongest difference in word use.
- In this example, the dominant pattern looks much more like time, leadership, and role than a clean party divide.



Wordfish model: words

When plotting estimated term weights and word fixed effects against each other this often produces an eiffel plot

- Words that are used often but do not help distinguish between documents: high fixed effects and low term weights
- Words that help distinguish between documents and are used less often: lower fixed effects and high term weights (in either direction)
- In the figure, labels are selected separately from both ends of `beta`, so we inspect language on each side of the scale.
- This can help with interpretation and



Validating Wordfish estimates

Validation of Wordfish is crucial because the estimated dimension is not labeled or guaranteed to be what you expect it to be.

Checklist for validating results:

- **Inspect term plots (e.g. Eiffel plot):** What are the most discriminating words? Do they reflect an ideological or thematic divide?
- **Read documents at both extremes (θ high and low):** Does the language used support your interpretation of the scale?
- **Compare to external benchmarks:** Use human-coded labels, expert surveys, or metadata (e.g., party affiliation).
- **Consider preprocessing effects (Denny & Spirling, 2018):** How sensitive are results to choices like stopword removal or stemming?

Remember: the model will always return a dimension – even when none exists.

Validating scaling models: Hjorth et al. (2015)

Assess how well Wordscores and Wordfish estimate party positions from manifestos.

- Applied to German and Danish party manifestos (1990s – 2010s)
- Compared model estimates to multiple external benchmarks:
  - **CHES expert surveys**
  - **Eurobarometer voter placements:** average self-placement of voters by party support
  - **CMP RILE index:** hand-coded ideological content
  - Explored the effects of:
  - Number of parties per election
  - Length and richness of manifestos
  - Cross-national variation

Table 1. Summary stats for German and Danish manifesto data.

	Germany	Denmark
Elections	9	24
Avg. manifestos per election	4.7	8.2
Avg. manifesto length (no. of words)	10,306	1,232.1
Std. dev. manifesto lengths	5,502.5	1,377.7

Source: Hjorth et al. (2015)

Danish manifestos: validation results

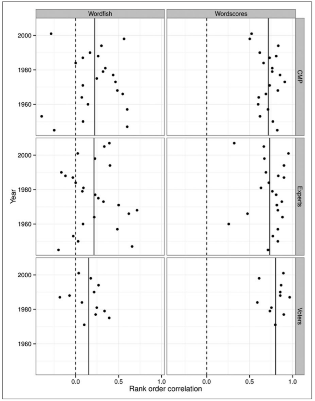


Figure 1. Wordfish and Wordscores estimates' rank order correlations with CMP, expert and voter estimates for each election year in the Danish sample. Vertical lines signify average rank order correlation across years.

Source: Hjorth et al. (2015)

German results

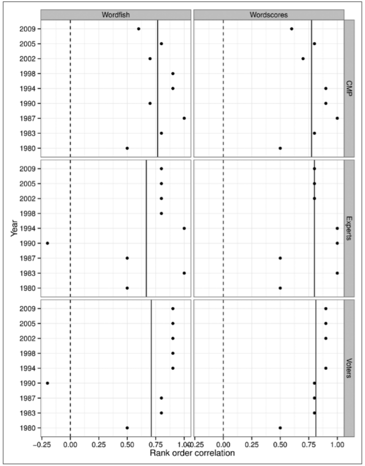


Figure 2. Wordfish and Wordscores estimates' rank order correlations with CMP, expert and voter estimates for each election year in the German sample. Vertical lines signify average rank order correlation across years.

Recommendations from Hjorth et al. (2015)

Guidelines based on text characteristics and available priors:

- Use Wordscores if:
  - You have **strong prior knowledge** about the ideological positions of reference texts
  - Your documents are relatively short or constrained (e.g., manifestos)
- Use Wordfish if:
  - You have **no strong priors**
  - Your documents are **long** and **ideologically polarized**
- If documents are **short** and **ideologically similar**:
  - Neither method may work well – consider gathering more data or switching methods

Note: These recommendations may need updating in light of newer semi-supervised methods (e.g., LSS), which combine domain knowledge with data-driven scaling.

Wordfish in quanteda

```
labour_anchor_index <- which(
  docvars(scaling_dfm, "party_family") == "Labour" &
    docvars(scaling_dfm, "leader") == labour_anchor_leader &
    docvars(scaling_dfm, "period") == wordscores_anchor_period
)

conservative_anchor_index <- which(
  docvars(scaling_dfm, "party_family") == "Conservative" &
    docvars(scaling_dfm, "leader") == conservative_anchor_leader &
    docvars(scaling_dfm, "period") == wordscores_anchor_period
)

hoc_wordfish <- textmodel_wordfish(
  scaling_dfm,
  dir = c(labour_anchor_index, conservative_anchor_index)
)
```

Wordfish among selected Conservative and Labour leaders

leader-election-period document	theta
Labour Jeremy Corbyn 2019–2024	-1.1805
Conservative Kemi Badenoch 2024–2025	-1.1760
Labour Keir Starmer 2019–2024	-1.1725
Labour Michael Foot 1979–1983	1.5259
Conservative Margaret Thatcher 1979–1983	2.4740
Conservative Margaret Thatcher 1983–1987	2.5543

- Here Wordfish appears to recover a strong period and leadership dimension.

Latent semantic scaling

Latent Semantic Scaling (LSS) is useful when we know something about the dimension, but do not want to hand-score reference documents.

$$\text{seed words} \rightarrow \text{nearby context words} \rightarrow \text{document scores}$$

- LSS combines dictionary logic with an embedding representation

LSS workflow

Semi-supervised method of scaling developed by Watanabe (2021).

What we provide

- A sentence-level corpus
- Seed words for the two poles of the scale
- A set of terms we want to score

What the model learns

- Which words occur in similar contexts
- Which terms are closer to the positive or negative seed pole
- How documents score after aggregating those term scores

The main shift from Wordscores: we no longer hand-score reference documents; we provide a small amount of conceptual guidance via seed words.

LSS in practice: economic language in HoC

When leaders talk about the economy, is the surrounding language more positive or negative?

1. Use sentiment seed words from `LSX`
2. Find terms near `econom*`, `tax*`, `budget*`, `inflation`, and `growth`
3. Estimate LSS on sentence-level HoC text
4. Inspect the high and low scoring terms before interpreting party scores

LSS code: the moving parts

```r
sentiment_seed <- as.seedwords(data_dictionary_sentiment)

economy_patterns <- c(
  "econom*", "tax*", "budget*", "inflation", "growth"
)

economy_terms <- char_context(
  lss_tokens,
  pattern = economy_patterns,
  p = 0.05
)

economy_terms <- economy_terms[economy_terms %in% featnames(lss_dfm)]
```

This narrows the analysis to words that appear near economic language. The next step estimates the scale using the seed words and those context terms.

Estimating and reading LSS

```r
hoc_lss <- textmodel_lss(
  lss_dfm,
  seeds = sentiment_seed,
  terms = economy_terms,
  k = 100,
  cache = FALSE
)
```

- `seeds`: the positive and negative poles of the scale
- `terms`: the economy-related context words to score
- `k`: the number of latent dimensions used internally

Always inspect the scored terms: otherwise the document scores are easy to overinterpret.

Instruction-based scaling

- LLMs make it tempting to score texts directly with a prompt:
  - "Place this speech from -1 to +1 on economic left–right."
  - "Return only a score and a one-sentence justification."
- This can be useful for exploration, codebook development, or candidate labels.
- It is not automatically a measurement model.
- Good practice:
  - define the scale before prompting;
  - evaluate against human-coded or metadata-based benchmarks;
  - inspect systematic errors by party, time period, speaker, and topic;
  - compare against other scaling methods

Instruction-based scaling: example prompt

Prompt sketch

You are coding UK parliamentary speeches. Place the speech on a one-dimensional economic left-right scale from -1 to $+1$.

- '-1' means the speech strongly emphasizes redistribution, public provision, labour protection, or economic intervention. - '$+1$' means the speech strongly emphasizes market competition, fiscal restraint, deregulation, or individual economic responsibility. - Return a numeric score, one sentence explaining the textual evidence, and 'uncertain $=$ TRUE/FALSE'.

Important: the prompt defines a measurement instrument. It must be validated.

Validating instruction-based scores

Check	Why it matters
Compare with human-coded speeches	tests whether the prompt captures the intended concept
Inspect high, low, and uncertain cases	reveals what textual evidence drives the scores
Compare across parties, periods, and agendas	detects systematic bias or role effects
Re-run with prompt variants or model variants	checks robustness
Record model, prompt, date, and settings	preserves provenance for replication